DATA COMMUNICATIONS & NETWORKS — QUIZ 2

Module: Lecture 1 Part 2 (Protocol Elements, OSI Layers & Network Scopes)

Name: _____ Student ID: _____ Score: _____ / 18

1. Complete the missing terminology in each of the following foundational statements:

(a) _____ refers to the exchange of digital data between two or more connected devices through a transmission medium.

(b) In network communications, a _____ refers to the agreed set of rules that govern data exchange, serving as the grammar and conventions of communication.

(c) _____ defines the structure, format, and arrangement of symbols/bits in a message.

(d) _____ defines the meaning and interpretation of each section of bits or field within a message.

2. Which OSI layer is directly responsible for transmitting and receiving unstructured, raw bits over a physical medium, including signal voltage levels, pin connectors, and data rates?

- ◯ a. Data Link Layer
- ◯ b. Physical Layer
- ◯ c. Transport Layer
- ◯ d. Network Layer
- ◯ e. Presentation Layer

3. Which OSI layer organizes raw bitstreams into *frames* and provides node-to-node hop delivery, media access control (MAC), flow control, and physical error detection?

- ◯ a. Session Layer
- ◯ b. Network Layer
- ◯ c. Data Link Layer
- ◯ d. Physical Layer
- ◯ e. Transport Layer

4. Which OSI layer handles logical addressing (e.g., IP addresses), routing, and packet forwarding across multiple heterogeneous networks from source to destination?

- ◯ a. Network Layer
- ◯ b. Data Link Layer
- ◯ c. Transport Layer
- ◯ d. Application Layer
- ◯ e. Presentation Layer

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5. Which transmission mode permits bidirectional communication between two stations, but *only in one direction at a time* (stations must take turns)?

- ◯ a. Simplex
- ◯ b. Full-duplex
- ◯ c. Half-duplex
- ◯ d. Multiplex
- ◯ e. Broadcast simplex

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6. A security camera streams real-time video back to a monitoring hub. When an operator sends a command to pan or zoom the lens, the video stream pauses momentarily until the movement command completes, after which video resumes. Which communication mode is demonstrated?

- ◯ a. Simplex, because the camera is primarily a sensor.
- ◯ b. Full-duplex, because both commands and video travel simultaneously.
- ◯ c. Half-duplex, because both devices can transmit, but not at the exact same instant.
- ◯ d. Asynchronous broadcast, because anyone can tap into the stream.
- ◯ e. Biplex, because commands and images are modulated on light.

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7. Which of the following is NOT an example of simplex communication?

- ◯ a. A commercial television station broadcasting over the air to household TVs.
- ◯ b. A desktop computer sending video signals to a standard computer monitor.
- ◯ c. A public train station PA system announcing departure gates to passengers.
- ◯ d. Two smartphones exchanging live conversational voice audio during a cell phone call.
- ◯ e. A traditional wireless keyboard transmitting keystrokes to a USB receiver dongle.

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8. Which sequence correctly arranges network classifications strictly in order from smallest to largest geographic coverage?

- ◯ a. $\mathrm{LAN} \rightarrow \mathrm{PAN} \rightarrow \mathrm{WAN} \rightarrow \mathrm{MAN}$
- ◯ b. $\mathrm{PAN} \rightarrow \mathrm{LAN} \rightarrow \mathrm{MAN} \rightarrow \mathrm{WAN}$
- ◯ c. $\mathrm{PAN} \rightarrow \mathrm{MAN} \rightarrow \mathrm{LAN} \rightarrow \mathrm{WAN}$
- ◯ d. $\mathrm{WAN} \rightarrow \mathrm{MAN} \rightarrow \mathrm{LAN} \rightarrow \mathrm{PAN}$
- ◯ e. $\mathrm{MAN} \rightarrow \mathrm{WAN} \rightarrow \mathrm{PAN} \rightarrow \mathrm{LAN}$

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9. Which statement best summarizes the primary engineering advantage of a layered network model (such as OSI or TCP/IP)?

- ◯ a. Dividing networks into layers eliminates all potential bit transmission errors.
- ◯ b. Hardware and software at any single layer can evolve and be upgraded independently without modifying other layers.
- ◯ c. Layering removes the necessity of implementing flow control or protocol formatting.
- ◯ d. Networks operate much faster because intermediate switches can skip lower layers entirely.
- ◯ e. It allows network applications to communicate without requiring physical cabling or wireless antennae.

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10. What are the three essential core elements that formally define any network protocol?

- ◯ a. Hardware, Software, and Firmware
- ◯ b. Syntax, Semantics, and Timing
- ◯ c. Copper, Fiber, and Radio
- ◯ d. Packets, Frames, and Bits
- ◯ e. Sender, Receiver, and Medium

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11. In the context of protocol elements, what does the Timing component govern?

- ◯ a. The physical pin arrangement of the Ethernet connector.
- ◯ b. The linguistic dictionary translation of the packet payload.
- ◯ c. When data should be sent and how fast it can be transmitted (speed matching & sequencing).
- ◯ d. The monetary billing rate per hour of connection.
- ◯ e. The exact GPS coordinates of the communicating devices.

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12. Match each Protocol Data Unit (PDU) name with its corresponding OSI Reference Model Layer:

(1) Segment / Datagram → OSI Layer: _____

(2) Packet → OSI Layer: _____

(3) Frame → OSI Layer: _____

(4) Raw Bits → OSI Layer: _____

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13. Which OSI layer is responsible for end-to-end process-to-process delivery, segmenting messages, assigning port numbers, and guaranteeing end-to-end reliability (e.g., TCP)?

- ◯ a. Physical Layer
- ◯ b. Network Layer
- ◯ c. Transport Layer
- ◯ d. Data Link Layer
- ◯ e. Session Layer

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14. A smartwatch measuring heart rate transmits data to a smartphone resting in the user's pocket via Bluetooth Low Energy. What network type best classifies this scenario?

- ◯ a. Local Area Network (LAN)
- ◯ b. Metropolitan Area Network (MAN)
- ◯ c. Personal Area Network (PAN)
- ◯ d. Wide Area Network (WAN)
- ◯ e. Global Area Network (GAN)

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15. A city administration interconnects municipal traffic cameras, fire departments, and public libraries across a 35-kilometer municipal district using a dedicated fiber ring. What network classification does this represent?

- ◯ a. Personal Area Network (PAN)
- ◯ b. Metropolitan Area Network (MAN)
- ◯ c. Local Area Network (LAN)
- ◯ d. Body Area Network (BAN)
- ◯ e. Storage Area Network (SAN)

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16. During the process of data transmission, when data moves down from the Application layer to the Physical layer, each layer attaches its own protocol control information. This process is called:

- ◯ a. Decapsulation
- ◯ b. Demultiplexing
- ◯ c. Encapsulation
- ◯ d. Modulation
- ◯ e. Quantization

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17. Walkie-talkie communication, where users must depress a Push-to-Talk (PTT) button and state "Over" before the other party can reply, is a classic real-world example of:

- ◯ a. Simplex communication
- ◯ b. Half-duplex communication
- ◯ c. Full-duplex communication
- ◯ d. Multiplexed concurrent communication
- ◯ e. Fully meshed communication

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18. Which OSI layer is responsible for translating data between disparate application formats, handling character encoding (such as ASCII or UTF-8), and providing data compression or encryption?

- ◯ a. Application Layer
- ◯ b. Session Layer
- ◯ c. Presentation Layer
- ◯ d. Transport Layer
- ◯ e. Network Layer

ANSWER KEY & DETAILED EXPLANATIONS

Lecture 1 Part 2 — Solutions

Question 1: Terminology Core Definitions

(a) Data Communication: Exchange of digital data between connected devices via a medium.
(b) Protocol: Agreed rules governing communication formatting, timing, and behavior.
(c) Syntax: The structure, arrangement, and bit format of message fields.
(d) Semantics: The specific meaning and contextual interpretation of each bit pattern.

Question 2: b OSI Layer 1

Explanation: The **Physical Layer** deals with the physical medium and transmission of raw bits. It defines signaling, voltage thresholds, line impedance, connector specs (RJ-45, fiber), and clock timings.

Question 3: c OSI Layer 2

Explanation: The **Data Link Layer** takes raw bits from the Physical layer and packages them into discrete structures called **frames**. It uses MAC (physical) addresses to ensure error-free transmission between neighboring nodes on the same local segment.

Question 4: a OSI Layer 3

Explanation: The **Network Layer** handles host-to-host delivery across independent networks. It provides logical addressing (IP addresses) and path determination (routing algorithms).

Question 5: c Directionality

Explanation: Half-duplex allows two-way communication, but only one party can transmit at any given instant. Simplex is strictly one-way; Full-duplex allows both parties to transmit simultaneously.

Question 6: c Communication Modes

Explanation: Because transmission occurs in both directions (video uplink from camera, PTZ control downlink to camera), but transmission pauses while the other transmits, it represents **Half-duplex** operation.

Question 7: d Simplex vs Duplex

Explanation: Smartphones during a telephone call use **Full-duplex** communication; both people can speak and hear each other at the same time over separated channels or frequency division. TV broadcasts, monitors, and public announcement speakers are purely one-way (simplex).

Question 8: b Network Coverage Hierarchy

Explanation: Arranged by geographic scale:

$$\text{PAN (meters)} \rightarrow \text{LAN (building/campus)} \rightarrow \text{MAN (city/metro area)} \rightarrow \text{WAN (country/globe)}$$

Question 9: b Modular Architecture

Explanation: The primary value of layered models is **modularity** and standard interfaces. An engineer can invent a new physical fiber technology (Layer 1) without modifying the web browser or HTTP (Layer 7).

Question 10: b Protocol Elements

Explanation: Every network communication protocol is defined by three fundamental elements:

- **Syntax:** Format and structure of the data.
- **Semantics:** Meaning of each section of bits.
- **Timing:** When data should be sent and how fast.

Question 11: c Protocol Timing

Explanation: Timing dictates synchronization between sender and receiver, preventing a fast sender from overwhelming a slow receiver (flow control) and avoiding collisions.

Question 12: PDU Identification OSI Model PDUs

(1) Segment / Datagram: Transport Layer (Layer 4)
(2) Packet: Network Layer (Layer 3)
(3) Frame: Data Link Layer (Layer 2)
(4) Raw Bits: Physical Layer (Layer 1)

Question 13: c OSI Layer 4

Explanation: The **Transport Layer** provides true end-to-end process-to-process delivery using port numbers (e.g., TCP port 80 or 443), performing segmentation, reassembly, and reliability controls.

Question 14: c PAN Classification

Explanation: Networks centered around a single person's immediate personal operating space (typically within a radius of 1 to 10 meters) are classified as **Personal Area Networks (PANs)**.

Question 15: b MAN Classification

Explanation: A **Metropolitan Area Network (MAN)** typically spans an entire town, municipality, or large city campus (typically 5 to 50 kilometers).

Question 16: c Encapsulation

Explanation: As application data moves down the stack, each successive layer prepends a header (and occasionally appends a trailer), wrapping lower-layer information. This process is called **encapsulation**.

Question 17: b Half-Duplex Real World

Explanation: Because the radio frequency channel is shared and only one person can broadcast at a time while the other listens, walkie-talkies are the quintessential example of **half-duplex** communication.

Question 18: c OSI Layer 6

Explanation: The **Presentation Layer** is responsible for data translation, character code conversion (e.g., ASCII, Unicode), data compression, and cryptographic encryption/decryption before passing data to the application.